

SAMRAT PAUL , PhD.

Curriculum Vitae – January 2026

Shishir Hostel., Indian Agricultural Research Institute (IARI), Pin: 110012, New Delhi, INDIA

(+91) 9614946594 | samrat16.sp@gmail.com | <https://github.com/SamratPaul16>ORCID [0009-0008-2203-7126](https://orcid.org/0009-0008-2203-7126)**RESEARCH OVERVIEW**

My Ph.D. research develops minimally invasive, AI-powered diagnostic protocols for plant disease surveillance, specializing in multiclass disease detection and pixel-level severity quantification through advanced image segmentation. I integrate deep learning architectures (e.g., YOLO, MobileNet, SAM) with attention-based enhancements for real-time field deployment, achieving simultaneous detection, symptom discrimination, and severity estimation while reducing manual assessment time by 12-fold. By combining thermal imaging for pre-symptomatic detection, RGB-based mobile disease indexing, CRISPR diagnostics, and epidemiological modeling, my work establishes non-destructive monitoring frameworks that bridge laboratory precision with scalable field implementation for sustainable, chemical-free management of plant pathogens.

EDUCATION**Indian Agricultural Research Institute**New Delhi | **2021-2026**

PhD in Plant Pathology, awarded 13 Feb 2026

Dissertation title: “Assessment of leaf curl infection risk in relation to host–vector interaction: aiming at the development of strategic management options”

Committee: Parimal Sinha (Advisor; Chair), Anirban Roy, Matiyar Rahaman Khan, Arpita Srivastava and Suresh Nebapure

Bidhan Chandra Krishi Viswavidyalaya,Mohanpur, WB | **2019-2021**

MSc in Plant Pathology

Uttarbanga Krishi Viswavidyalaya,Pundibari, WB | **2015-2019**

BSc in Agriculture

WORK & RESEARCH EXPERIENCE**Doctoral Research Assistant**IARI, New Delhi | **2021-2026**

Dr. Parimal Sinha, Indian Agricultural Research Institute, Division of Plant Pathology, Epidemiology

Dr. Anirban Roy, Indian Agricultural Research Institute, Division of Plant Pathology, Advanced center of Plant Virology

My doctoral research focused on integrated management of chili leaf curl virus, with an emphasis on computer vision-based disease detection and differentiation of different leaf curling patterns, molecular diagnostics, and epidemiological modeling based on vector feeding and migration rates.

- Developed novel machine learning architectures for non-invasive disease surveillance: SCA-MobiPlant classifier and YOLOv9t-DyE object detector integrating MobileSAM for pixel-level severity quantification
- Integrated thermal imaging for pre-symptomatic detection with SAM-based automated canopy segmentation, eliminating manual annotation bottlenecks in phenotyping workflows
- Established epidemiological models using whitefly contact rate dynamics, identifying optimal

physical barrier intervention period

- Deployed Android application integrating real-time disease detection, incidence calculation, and field surveillance capabilities
- Engineered RPA-CRISPR-Cas12a lateral flow assay for rapid ChiLCV detection from crude leaf extracts

Graduate Research Assistant (M.Sc.)

BCKV, WB | 2019-2021

Dr. Sujit Kumar Roy, Bidhan Chandra Krishi Viswavidyalaya, Department of Plant Pathology, Bacteriology

My master's research focused on biocontrol strategies for cowpea diseases using bacterial consortia, with an emphasis on plant growth promotion, disease suppression, and sustainable yield enhancement.

- Evaluated synergistic effects of *Bacillus subtilis* and *Pseudomonas aeruginosa* consortia on cowpea growth parameters, demonstrating enhanced shoot/root development and maximum nodulation compared to individual strain applications
- Established field efficacy of bacterial consortia for web blight suppression and sustainable yield improvement in cowpea production systems
- Optimized cost-effective mass production protocols for *Bacillus* sp. B-11 using nutrient-enriched organic substrates

Research Assistant, Dr. Matiyar Rahaman Khan

IARI, New Delhi | 2024-2025

Division of Nematology

Research Assistant, Dr. Subrata Dutta

BCKV, WB | 2019-2021

Department of Plant Pathology

Undergraduate Biocontrol Project

2018

PEER-REVIEWED PUBLICATIONS

- Paul, S.**, Emmadi, V., Sarkar, M., Das, S., Roy, A., & Sinha, P. (2025). SCA-MobiPlant: smartphone-deployed multistage attention fusion model for accurate field detection of chili leaf curl complex. *Plant Methods*, 21(1), 138. <https://doi.org/10.1186/s13007-025-01453-x>
- Paul, S.**, Das, S., Khan, M. R., Srivastava, A., Nebapure, S., Roy, A., & Sinha, P. (2025). YOLOv9t-DyE: A lightweight detection framework with SAM-assisted segmentation for quantifying chilli leaf curl complex. *Smart Agricultural Technology*, 6, Article 101764. <https://doi.org/10.1016/j.atech.2025.101764>
- Paul, S.**, Emmadi, V., Saxena, S., Sarkar, M., Mandal, B., Kumar, R., ... & Roy, A. (2025). RPA-assisted CRISPR-Cas12a-Enabled Point-of-Care Diagnostic Platform for Chilli Leaf Curl Virus with Fluorescent and Colorimetric Readouts. *Frontiers in Microbiology*, 16, 1644322. <https://doi.org/10.3389/fmicb.2025.1644322>
- Chavan, D. D., **Paul, S.**, Thapa, P., Madival, S. D., Sharma, S. K., Saha, S., ... & Roy, A. (2025). Integrated Transcriptomics and Targeted Metabolomics Delineate the Mechanisms of MYMIV Resistance in Soybean Via Regulation of Isoflavonoid Biosynthesis, Hormone Signalling, and RNA Silencing Pathways. *Journal of Plant Growth Regulation*, 1-24. <https://doi.org/10.1007/s00344->

[025-11992-1](#)

- Sarkar, M., Gupta, D., Singh, O. W., **Paul, S.**, Kumar, R., Mandal, B., & Roy, A. (2024). Identification of pathogenicity determinants in ToLCNDV and their RNAi-based knockdown for disease management in *Nicotiana benthamiana* and tomato plants. *Frontiers in Microbiology*, 15, 1481523. <https://doi.org/10.3389/fmicb.2024.1481523>
- Manojkumar C, Limbola M, **Paul S**, et al. CRISPR-Cas12a–based lateral flow detection of white spot syndrome virus: a dual-target approach for detection of early and latent infection. *Journal of Veterinary Diagnostic Investigation*. 2026;0(0). doi:[10.1177/10406387251399147](https://doi.org/10.1177/10406387251399147)
- Paul, S.**, Pati, S., Patwari, A., Dutta, S., Mandal, R., & Roy, S.K. (2024). *Plant growth promotion and suppression of web blight disease of cowpea by native rhizobacterial strains of Bacillus subtilis in combination with Pseudomonas aeruginosa*. *Biological Forum – An International Journal*, 16(1), 274–281.
- Pati, S., Patwari, A., **Paul, S.**, Kumar, S.K., Kumar, R., Barman, A. R., ... & Dutta, S. (2024). Mass multiplication of native rhizospheric bacteria on cheaply available organic substrates and their plant growth promotion potentiality on cowpea. *Ecology, Environment & Conservation*, 30, S291-S300. <http://doi.org/10.53550/EEC.2024.v30i03s.051>

IN-REVIEW PUBLICATIONS

- Das, B., Venu, E., Ramya, A., Vimalkumar, C., Sathiyaseelan, K., **Paul, S.**, Roy, A., & Sinha, P. *Geospatial modeling and risk prediction of chili leaf curl disease in India under changing climatic conditions* (Manuscript under review). *Discover Applied Sciences*, (Springer Nature).
- Chavan, D. D., **Paul, S.**, Thapa, P., Madival, S. D., Sharma, S. K., Saha, S., ... Roy, A. (2026). *Exploration of antioxidant and biochemical profile of Glycine max L. Merr. (soybean) in response to mungbean yellow mosaic India virus infection* (Manuscript under review). *Journal of Plant Growth Regulation*. (Springer Nature).

OTHER PUBLICATIONS

- Paul, S.** (2025). Plant disease diagnosis: Methods for identification of plant pathogens. In *Plant pathology diagnosis and management of crop disease* (Chapter 4, pp. 41–63). Nitya Publications.
- Paul, S.**, Dash, S. S., & Kumar, R. (2025). Smart technologies and digital innovations in horticulture. In *Innovative practices in sustainable horticulture* (Vol. I, Chapter 4, pp. 86–105). Ninetales Publishings.
- Paul, S.**, & Sahoo, B. C. (2025, March). Occupational health hazards of workers: Application of computer vision in plant disease screening. *AgriGate—An International Multidisciplinary e-Magazine*, 478–452. <https://doi.org/10.13140/RG.2.2.32637.47842>
- Paul, S.**, & Sahoo, B. C. (2025, March). CRISPR-Cas systems in molecular diagnostics for plant pathogens. [Agriculture & Food: E-News Letter], Article ID 70391.
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Patents & Copyright

- Patent: Roy, A., Sinha, P., Mandal, B., **Paul, S.**, Emmadi, V., & Saxena, S., A Rapid, Sensitive, Specific and Field Deployable Diagnostic Process For Chilli Leaf Curl Virus. Patent application filed (India), 2024
- Copyright: **Paul, S.**, Roy, A., & Sinha, P., Android Application: Digital surveillance system for monitoring leaf curl disease in Chilli. Copyright registered, 2025

PRESENTATIONS

- Paul, S.**, (17 Oct 2025) Assessment of leaf curl infection risk in relation to host–vector interaction: aiming at the development of strategic management options. PhD defense seminar.
- Paul, S.**, Roy, A., & Sinha, P. (8 Dec 2025) Next-generation plant health: AI-powered mobile detection with CRISPR validation for chili leaf curl management. Indian Council of Medical Research & National Institute of Virology, International Conference on “Changing Landscapes in Human, Animal and Plant Viruses: Bridging Basic Science, Innovation and Public Health”, VIROCON-2025.Oral Presentation. Nagpur, Maharashtra.
- Paul, S.**, Emmadi, V., Sinha, P., Mandal, B., & Roy, A. (1 Feb 2024) Recombinase polymerase amplification coupled CRISPR-Cas12a-based lateral flow assay for on spot detection of chilli leaf curl virus. Indian Phytopathological Society, National Conference on “Plant Health for Food Security: Threats and Promises”, ICAR-Indian Institute of Sugarcane Research (IISR). Poster session. Lucknow, Uttar Pradesh.
- Paul, S.**, Roy, A., & Sinha, P. (15 Jan 2025) Integrated Framework for Rapid Field Detection and Confirmation of Chili Leaf Curl Disease Using Deep Learning and CRISPR-Cas12a Diagnostics. Indian Phytopathological Society, International Conference on “Integrating Genomics in Plant Pathology: New Frontiers in Disease Management”, ICAR-Indian Agricultural Research Institute (IARI). MJ Narasimhan Academic Merit Award Contest . New Delhi.
- Paul, S.**, Pati, S., Patwari, A., Dutta, S., Mandal, R., & Roy, S.K. (6 Feb 2022) Influence of native rhizobacterial *Bacillus* sp. disease suppression of cowpea. Indian Phytopathological Society-East zone cum National symposium, Bidhan Chandra Krishi Viswavidyalaya (BCKV).). Poster session. Mohanpur, West Bengal.

Training Program Participated

- Data Analysis with R** — Centre for e-Learning, Kerala Agricultural University (with Dept. of Agricultural Statistics) (Aug 7–11, 2023)
- Data Science and Agriculture (Online Certificate Course)** — DST-TEC, G.B. Pant University of Agriculture & Technology
(3-month program, Aug–Nov 2024;) Grade: A+
- Python Programming (Training Program)** — ICAR–Indian Agricultural Statistics Research Institute (IASRI), New Delhi
(Aug 2–8, 2024)

TEACHING

- Teaching Assistant**, Advanced Virology (Practical) (PL-PATH 602) **Semester I, 2024**
Division of Plant Pathology, ICAR-Indian Agricultural Research Institute
Principal Scientist: Dr. Anirban Roy

MENTORSHIP

- Maharshi Limbola**, Research Associate, ICFRE **2025**
Project: Experimental design and execution of CRISPR-Cas12a–based lateral flow assay for viral diagnostics
Products: Co-author, *Journal of Veterinary Diagnostic Investigation* (2026) — CRISPR-Cas12a–based lateral flow detection of white spot syndrome virus
- Dharmappa D. Chavan**, Doctoral Researcher, IARI **2025**
Project: Integrated transcriptomics and metabolomics to dissect MYMIV resistance mechanisms in soybean

Mentorship Focus: Statistical analysis, data interpretation, and multi-omics integration
Products: Co-author, *Journal of Plant Growth Regulation* (2025)

Mehulee Sarkar, Doctoral Researcher, IARI

2024

Project: Identification of pathogenicity determinants in ToLCNDV and RNAi-based functional validation

Mentorship Focus: Statistical analysis and experimental data interpretation

Products: Co-author, *Frontiers in Microbiology* (2024)

Susmita Pati, Graduate Researcher, BCKV

2023

Project: Field experimentation and evaluation of plant growth-promoting effects of native rhizospheric bacteria on cowpea

Mentorship Focus: Field trial execution, data collection, and experimental standardization

Products: Co-author, *Ecology, Environment & Conservation* (2024)

TECHNICAL SKILLS

Wet Lab & Molecular Diagnostics

PCR, qPCR, RPA, CRISPR–Cas12a–based detection, ELISA, nucleic acid extraction (DNA/RNA), viral diagnostics, lateral flow assay development,

Field & Experimental Platforms

Plant disease assessment, vector population monitoring, epidemiological field surveys, controlled environment experiments, multi-location field trials, sampling strategy optimization

Programming Languages

Python, R,

Frameworks & Libraries

TensorFlow, PyTorch, OpenCV, scikit-learn, Pandas, NumPy

Machine Learning & Computer Vision

Deep learning (CNNs), object detection and classification, image segmentation, thermal image analysis, model deployment,

Epidemiological & Statistical Modeling

Population dynamics, survival analysis, host–pathogen–vector interaction modeling, disease severity quantification

Tools & Platforms

Google Colab, GitHub, Android Studio, TensorFlow, VS Code

Data Analysis & Visualization

Statistical analysis, multi-omics data interpretation, data visualization, reproducible workflows

AWARDS

Best Oral Presentation Award (2025). Indian Council of Medical Research & National Institute of Virology, International Conference on “Changing Landscapes in Human, Animal and Plant Viruses: Bridging Basic Science, Innovation and Public Health”, VIROCON-2025.

Best Poster Presentation Award (2024). Indian Phytopathological Society, National Conference on “Plant Health for Food Security: Threats and Promises”, ICAR-Indian Institute of Sugarcane Research (IISR).

ICAR-IARI PhD Merit Medal winner (2026). Thesis title “Assessment of leaf curl infection risk in relation to host–vector interaction: aiming at the development of strategic management options”

Shashya Suraksha Best PhD Thesis Award (2026) for the dissertation titled “Assessment of leaf curl infection risk in relation to host–vector interaction: aiming at the development of strategic management options” in the International Conference on “Advances in Digital Solutions and Green Technologies in Crop Pest Management” organized by AAPP & Agriculture and ecological research unit, ISI, Kolkata.

PROFESSIONAL MEMBERSHIPS

Indian Phytopathological Society

2022-Present